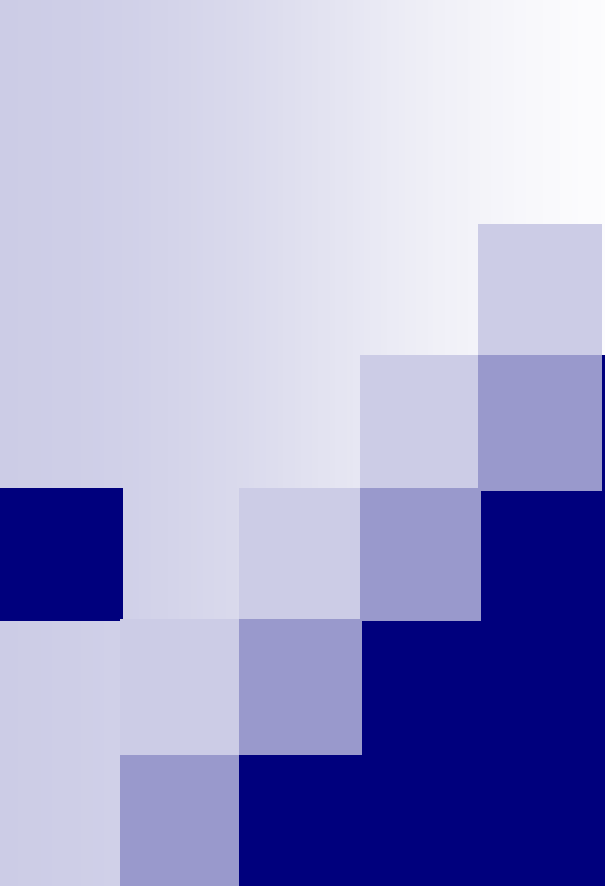
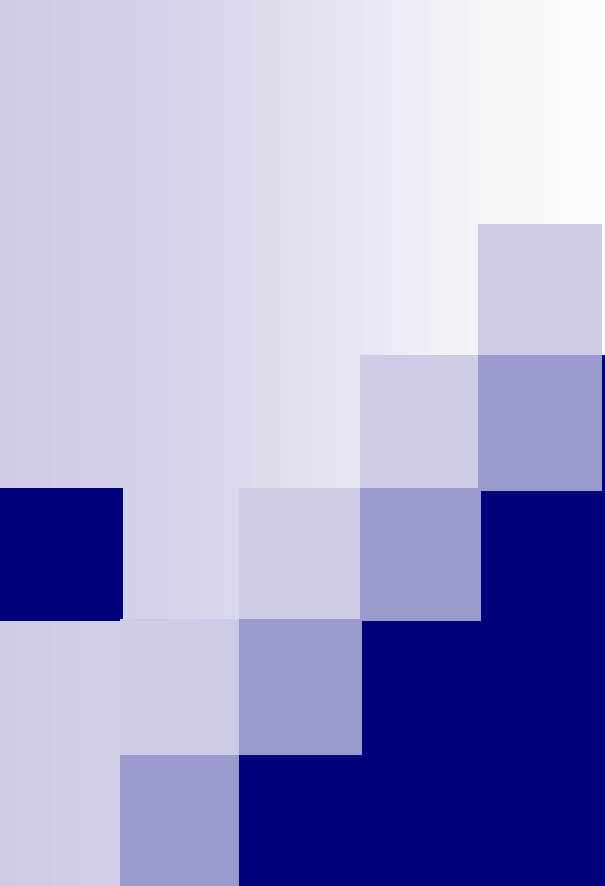


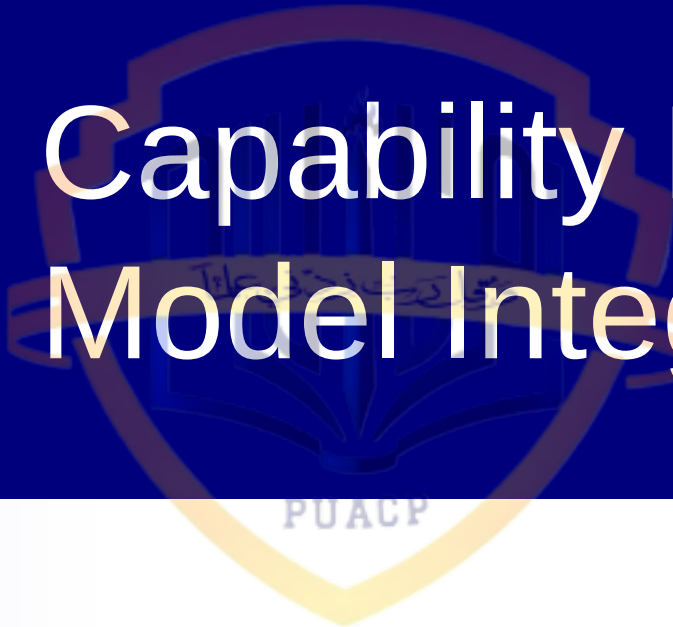


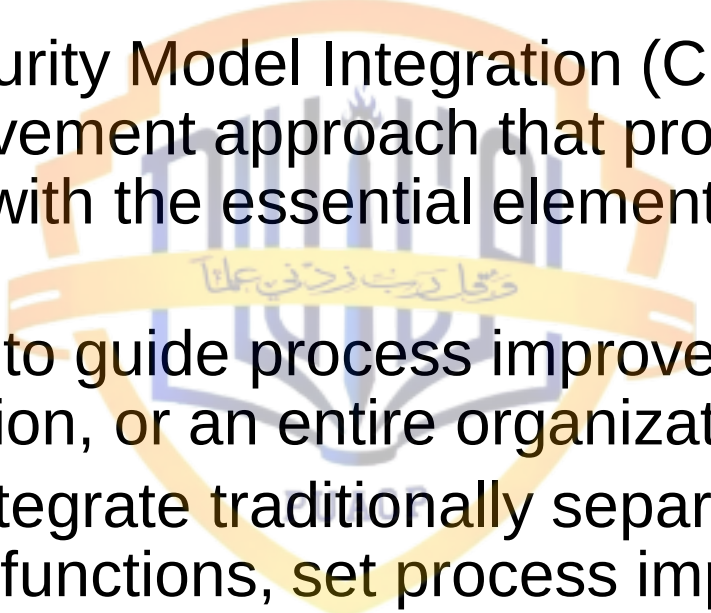
Process Management and Improvement – 2 - CMMI

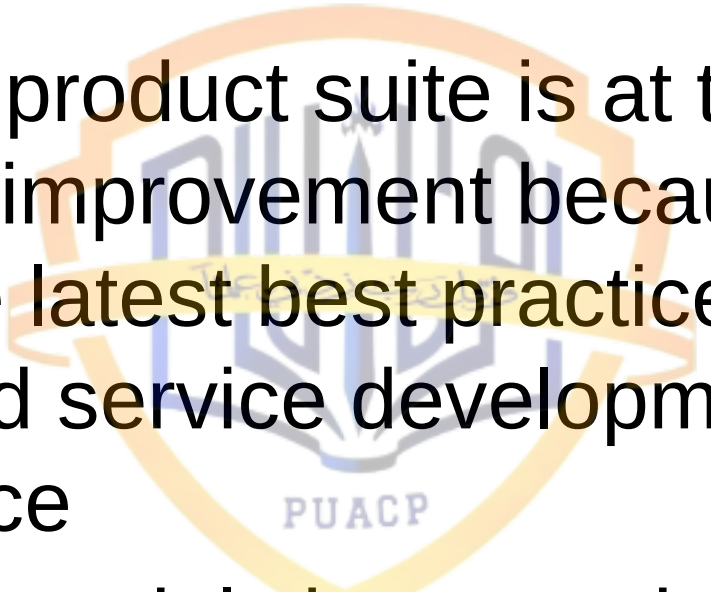
Lecture # 40



Capability Maturity Model Integration



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- Capability Maturity Model Integration (CMMI) is a process improvement approach that provides organizations with the essential elements of effective processes
 - It can be used to guide process improvement across a project, a division, or an entire organization
 - CMMI helps integrate traditionally separate organizational functions, set process improvement goals and priorities, provide guidance for quality processes, and provide a point of reference for appraising current processes

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- The CMMI product suite is at the forefront of process improvement because it provide the latest best practices for product and service development and maintenance
 - The CMMI models improve the best practices of previous models in many important ways

Benefits of CMMI - 1

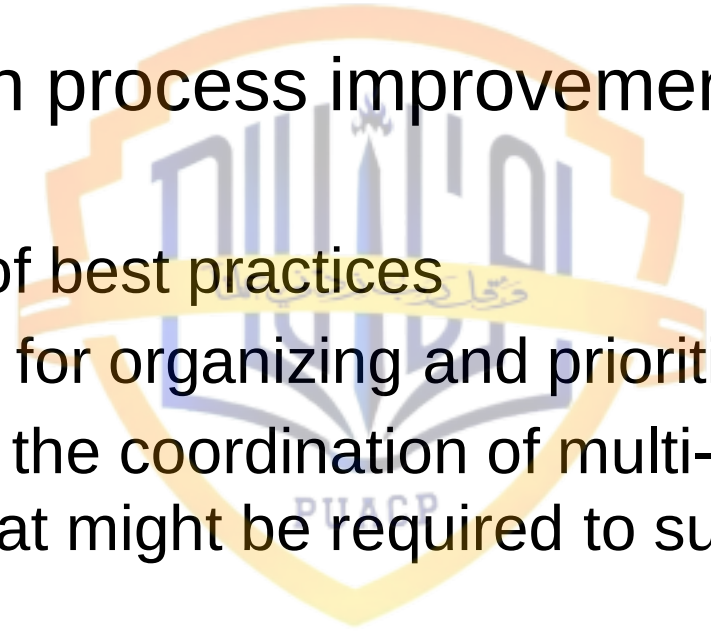
- CMMI best practices enable organizations to do the following
 - More explicitly link management and engineering activities to their business objectives
 - Expand the scope of and visibility into the product lifecycle and engineering activities to ensure that the product or service meets customer expectations

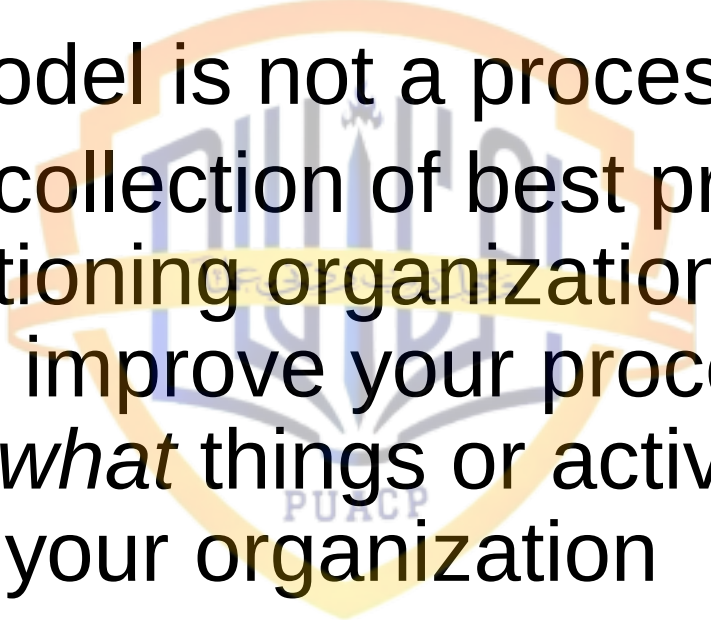
Benefits of CMMI - 2

- Incorporate lessons learned from additional areas of best practices (e.g., measurement, risk management, and supplier management)
- Implement more robust high-maturity practices
- Address additional organizational functions critical to their products and services
- More fully comply with relevant ISO standards

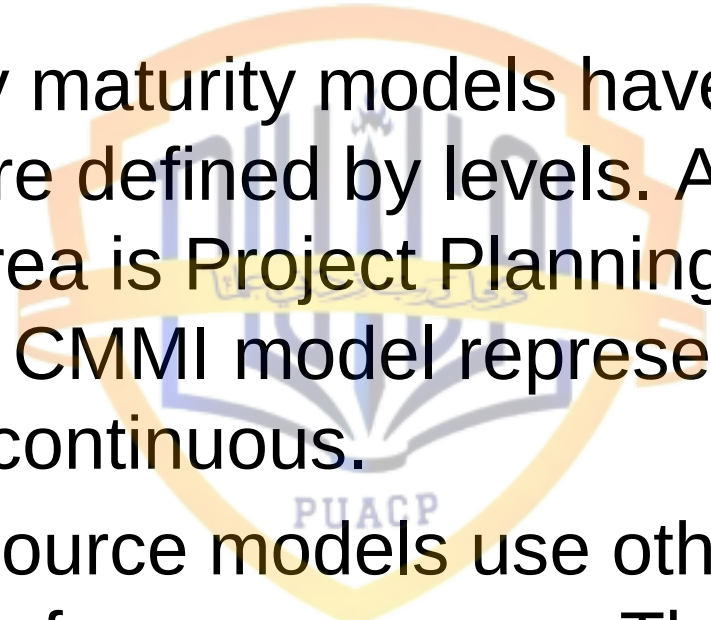
Background of CMMI

- CMM-SW: CMM Software
- SE CMM: System Engineering CMM
- SA CMM: Software Acquisition CMM
- IPD CMM: Integrated Product Development Team CMM
- System Engineering Capability Assessment Model (SECAM)
- System Engineering Capability Model (SECM)

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- Use CMMI in process improvement activities as a
 - Collection of best practices
 - Framework for organizing and prioritizing activities
 - Support for the coordination of multi-disciplined activities that might be required to successfully build a product
 - Means to emphasize the alignment of the process improvement objectives with organization business objectives

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- A CMMI model is not a process
 - CMMI is a collection of best practices from highly functioning organizations collected to help you improve your processes by describing *what* things or activities should be done in your organization
 - A CMMI model describes the characteristics of effective processes

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- All of the source models for CMMI are considered capability maturity models; however, each has a different approach. Review and examination of each source model led to the discovery of two types of approaches to presenting capability maturity models. These types of approaches have been given the label "representations" in the process improvement community. A representation reflects the organization, use, and presentation of components in a model

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- All capability maturity models have process areas that are defined by levels. An example of a process area is Project Planning. There are two types of CMMI model representations: staged and continuous.
 - Two of the source models use other terms for the concept of a process area. The Software CMM uses the term key process areas; the SECM uses the term focus areas.

Staged Representation

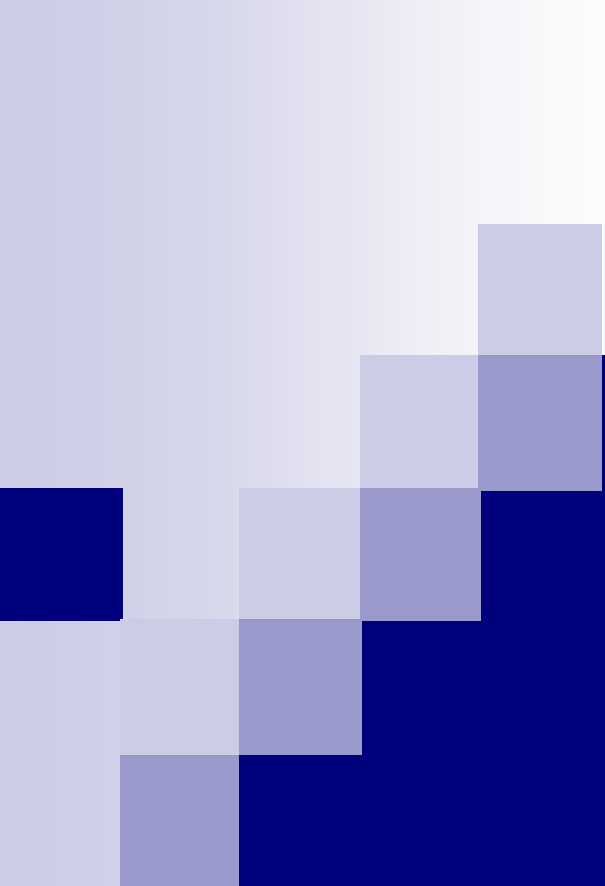
- The staged representation is the approach used in the Software CMM. It is an approach that uses predefined sets of process areas to define an improvement path for an organization. This improvement path is described by a model component called a maturity level
- A maturity level is a well-defined evolutionary plateau toward achieving improved organizational processes

Continuous Representation

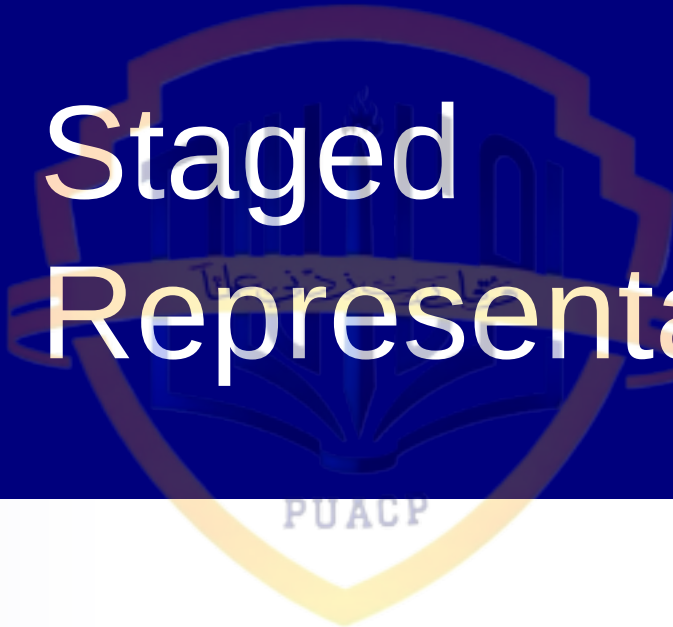
- The continuous representation is the approach used in the SECM and the IPD-CMM. This approach allows an organization to select a specific process area and improve relative to it
- The continuous representation uses capability levels to characterize improvement relative to an individual process area

CMMI Model Structure

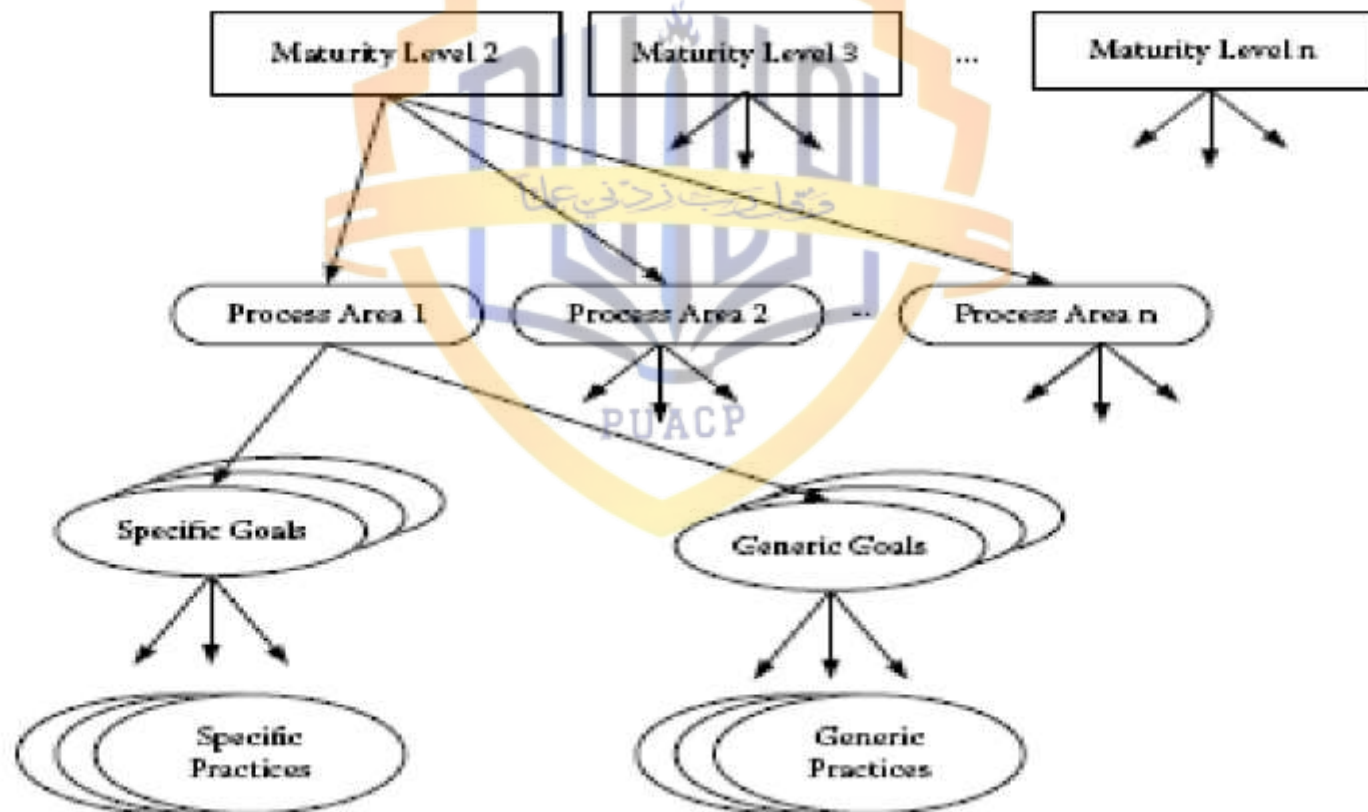
- Maturity Levels (staged representation) or Capability Levels (continuous representation)
- Process Areas
- Goals – Generic and Specific
- Practices – Generic and Specific



Staged Representation



CMMI Model Components in the Staged Representation



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- The staged representation offers a systematic, structured way to approach process improvement one step at a time. Achieving each stage ensures that an adequate improvement has been laid as a foundation for the next stage

- Process areas are organized by maturity levels that take much of the guess work out of process improvement. The staged representation prescribes the order for implementing each process area according to maturity levels, which define the improvement path for an organization from the initial level to the optimizing level. Achieving each maturity level ensures that an adequate improvement foundation has been laid for the next maturity level and allows for lasting, incremental improvement

Maturity Level

- Maturity level signifies the level of performance that can be expected from an organization
- There are five maturity levels

Maturity Levels

- Adhoc
- Managed
- Defined
- Quantitatively Managed
- Optimizing



Process Areas

- Each maturity level consists of several process areas
- A process area is a group of practices or activities performed collectively in order to achieve a specific objective

Goals

- Each PA has several goals that need to be satisfied in order to satisfy the objectives of the PA. There are two types of goals:
 - Specific goals (SG): goals that relate only to the specific PA under study
 - Generic goals (GG): goals that are common to multiple process areas throughout the model. These goals help determine whether the PA has been institutionalized

Practices

- Practices are activities that must be performed to satisfy the goals for each PA. Each practice relates to only one goal. There are two types of practices:
 - Specific practices (SP): practices that relate to specific goals
 - Generic practices (GP): practices associated with the generic goals for institutionalization

Level 1: Adhoc



Level 2: Defined - 1

- Adhering to organizational policies
- Following a documented plan and process description
- Applying adequate funding and resources
- Maintaining appropriate assignment of responsibility and authority
- Training people in their appropriate processes
- Placing work products under appropriate configuration management

Level 2: Defined - 2

- Monitoring and controlling process performance, and taking corrective action
- Objectively reviewing the process, work products, and services, and addressing noncompliance
- Reviewing the activities, status, and results of the process with appropriate levels of management, and taking corrective action
- Identifying and interacting with relevant stakeholders

Level 3: Managed - 1

- The organization has achieved all of the goals of Level 2. There is an organizational way of doing business, with tailoring of this organizational method allowed under predefined conditions. The organization has an organization's set of standard processes (OSSP)

Level 3: Managed - 2

- The following characteristics of the process are clearly stated
 - ☐ Purpose
 - ☐ Inputs
 - ☐ Entry criteria
 - ☐ Activities
 - ☐ Roles
 - ☐ Measures
 - ☐ Verification steps
 - ☐ Outputs
 - ☐ Exit criteria



Level 3: Managed - 3

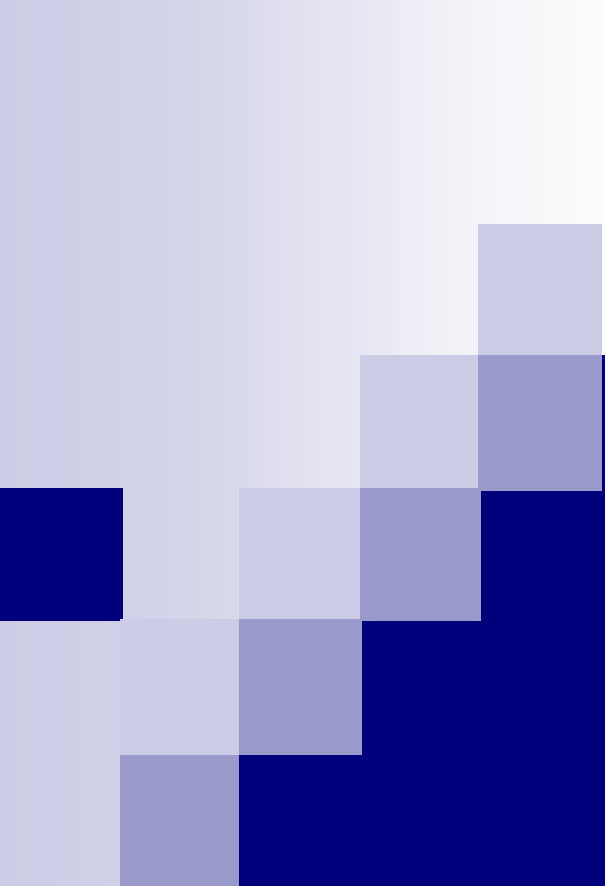
- Level 3 continues with defining a strong, meaningful, organization-wide approach to developing products. An important distinction between Level 2 and Level 3 is that at Level 3, processes are described in more detail and more rigorously than at Level 2. Processes are managed more proactively, based on a more sophisticated understanding of the interrelationships and measurements of the processes and parts of the processes. Level 3 is more sophisticated, more organized, and establishes an organizational identity—a way of doing business particular to this organization

Level 4: Quantitatively Managed

- The organization controls its processes by statistical and other quantitative techniques. Product quality, process performance, and service quality are understood in statistical terms and are managed throughout the life of the processes

Level 5: Optimizing

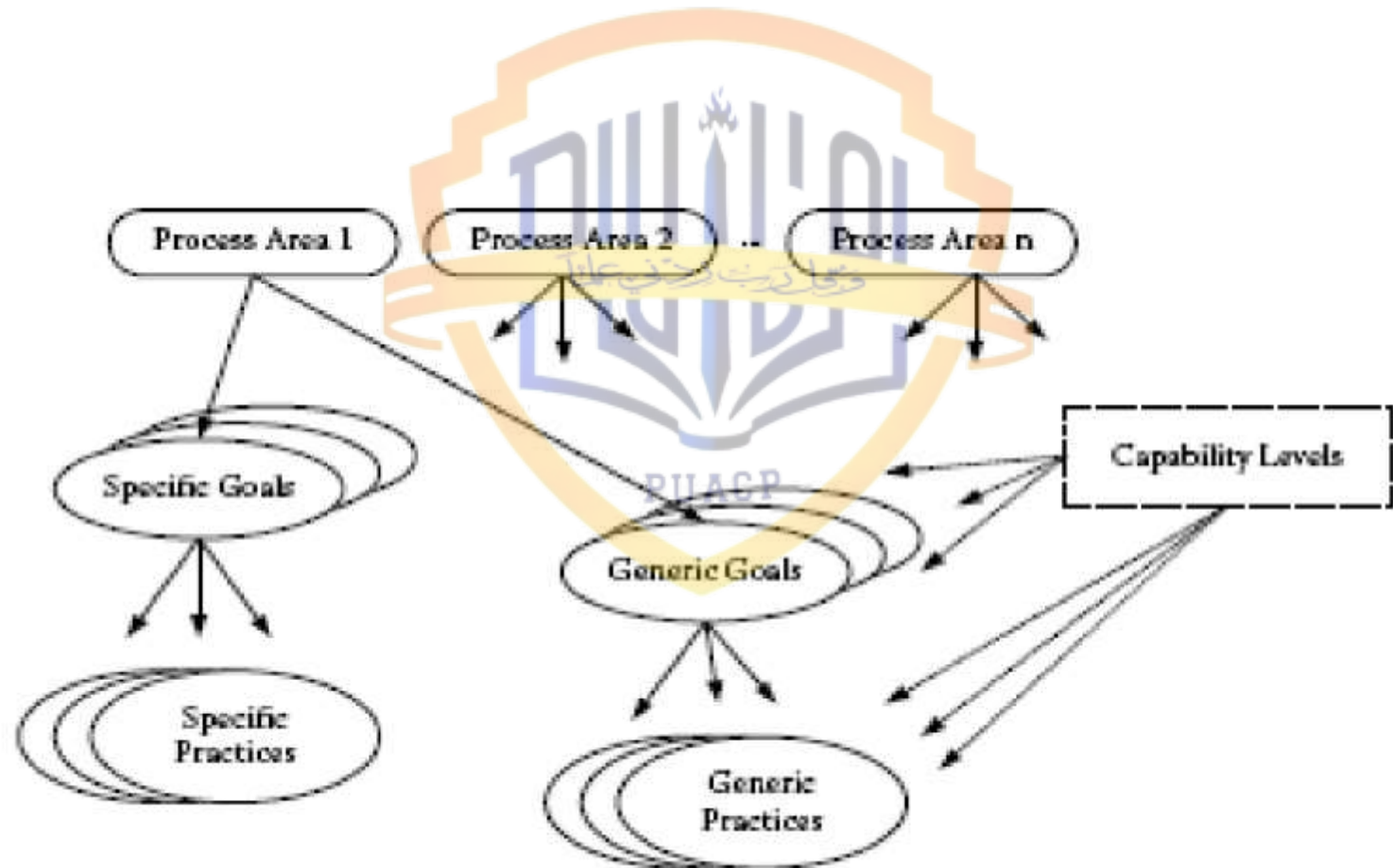
- Processes are continually improved based on an understanding of *common* causes of variation within the processes
- Level 5 is nirvana
- Everyone is a productive member of the team, defects are reduced, and your product is delivered on time and within the estimated budget



Continuous Representation



CMMI Model Components in the Continuous Representation



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- The continuous representation offers a flexible approach to process improvement. An organization may choose to improve the performance of a single process-related trouble spot, or it can work on several areas that are closely aligned to the organization's business objectives. The continuous representation also allows an organization to improve different processes at different rates. There are some limitations on an organization's choices because of the dependencies among some process areas

Continuous Representation

- The continuous representation uses the same basic structure as the staged representation.
- However, each PA belongs to a Process Area Category. A Process Area Category is just a simple way of arranging PAs by their related, primary functions

- Capability levels are used to measure the improvement path through each process area from an unperformed process to an optimizing process. For example, an organization may wish to strive for reaching capability level 2 in one process area and capability level 4 in another. As the organization's process reaches a capability level, it sets its sights on the next capability level for that same process area or decides to widen its scope and create the same level of capability across a larger number of process areas

Process Area Categories

- Process management
- Project management
- Engineering
- Support



Process Management

- Organizational Process Focus
- Organizational Process Definition (with Integrated Product and Process Development—IPPD)
- Organizational Training
- Organizational Process Performance
- Organizational Innovation and Deployment

Project Management

- Project Planning
- Project Monitoring and Control
- Supplier Agreement Management
- Integrated Project Management (with Integrated Product and Process Development—IPPD)
- Risk Management
- Quantitative Project Management

Engineering

- Requirements Development
- Requirements Management
- Technical Solution
- Product Integration
- Verification
- Validation
- (listed in increasing order of complexity)

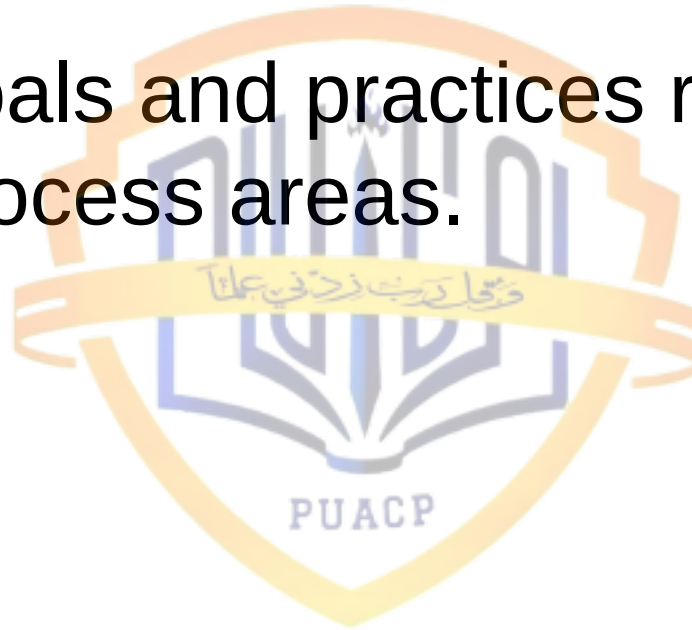
Support

- Configuration Management
- Process and Product Quality Assurance
- Measurement and Analysis
- Decision Analysis and Resolution
- Causal Analysis and Resolution

Goals and Practices

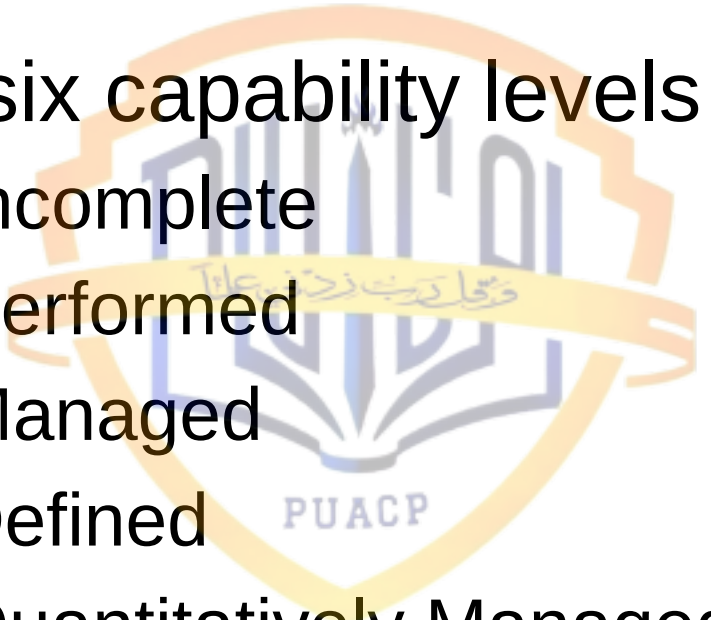
- Specific goals and practices relate to specific process areas and relate to tasks that make sense for that process area only. For example, Project Planning requires a project plan. Quantitative Project Management requires a process performance baseline

- Generic goals and practices relate to multiple process areas.



- CMMI focuses on institutionalization. Goals cannot be achieved without proving institutionalization of the process. Generic goals and generic practices support institutionalization and increasing sophistication of the process. Specific goals and specific practices support implementation of the process area. Process maturity and capability evolve. Process improvement and increased capability are built in stages because some processes are ineffective when others are not stable

- The continuous representation has the same basic information as the staged representation, just arranged differently; that is, in capability levels not maturity levels, and process area categories. The continuous representation focuses process improvement on actions to be completed within process areas, yet the processes and their actions may span different levels. More sophistication in implementing the practices is expected at the different levels. These levels are called capability levels

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- There are six capability levels
 - Level 0: Incomplete
 - Level 1: Performed
 - Level 2: Managed
 - Level 3: Defined
 - Level 4: Quantitatively Managed
 - Level 5: Optimizing

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- What's a capability level? Capability levels focus on maturing the organization's ability to perform, control, and improve its performance in a process area. This ability allows the organization to focus on specific areas to improve performance of that area

Level 0: Incomplete

- An incomplete process does not implement all of the Capability Level 1 specific practices in the process area that has been selected. This is tantamount to Maturity Level 1 in the staged representation

Level 1: Performed

- A Capability Level 1 process is a process that is expected to perform all of the Capability Level 1 specific practices. Performance may not be stable and may not meet specific objectives such as quality, cost, and schedule, but useful work can be done
- This is only a start, or baby step, in process improvement. It means you are doing something, but you cannot prove that it is really working for you

Level 2: Managed

- A managed process is planned, performed, monitored, and controlled for individual projects, groups, or stand-alone processes to achieve a given purpose. Managing the process achieves both the model objectives for the process as well as other objectives, such as cost, schedule, and quality. As the title of this level states, you are actively managing the way things are done in your organization. You have some metrics that are consistently collected and applied to your management approach

Level 3: Defined

- A defined process is a managed process that is tailored from the organization's set of standard processes. Deviations beyond those allowed by the tailoring guidelines are documented, justified, reviewed, and approved. The organization's set of standard processes is just a fancy way of saying that your organization has an identity. That is, there is an organizational way of doing work that differs from the way another organization within your company may do it

Level 4: Quantitatively Managed

- A quantitatively managed process is a defined process that is controlled using statistical and other quantitative techniques. Product quality, service quality, process performance, and other business objectives are understood in statistical terms and are controlled throughout the life cycle

Level 5: Optimizing

- An optimizing process is a quantitatively managed process that is improved based on an understanding of the common causes of process variation inherent in the process. It focuses on continually improving process performance through both incremental and innovative improvements. Both the defined processes and the organization's set of standard processes are targets of improvement activities

Summary



References

- <http://www.sei.cmu.edu/general/index.html>
- Interpreting the CMMI: A Process Improvement Approach, Second Edition, by Margaret K. Kulpa and Kent A. Johnson, Auerbach Publication, 2008 (electronic file), (Chapter 1-4)